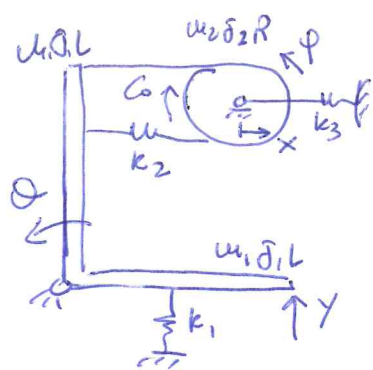


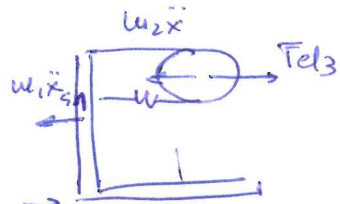


$$x = \begin{Bmatrix} \theta \\ \dot{x} \end{Bmatrix}$$

$$\Delta u = \begin{bmatrix} L/2 & 0 \\ 1 & 0 \\ 0 & 1 \\ L/R & 1/R \end{bmatrix} \begin{matrix} 2m_1 \\ 2F_1 \\ m_2 \\ \bar{g}_2 \end{matrix}$$

$$\Delta k = \begin{bmatrix} L/2 & 0 \\ \frac{3}{2}L & 2 \\ 0 & -1 \end{bmatrix} \quad \Delta \phi = \begin{bmatrix} -L/R & -1/R \end{bmatrix}$$

$$K_s = \begin{bmatrix} -m_1 g L/2 & 0 \\ 0 & 0 \end{bmatrix}$$



$$H_0 = m_1 \ddot{x}_{c1} + m_2 \ddot{x} - \bar{T} e_3$$

H_0

$$x = \begin{Bmatrix} \theta \\ \phi \end{Bmatrix} \quad \Delta u = \begin{bmatrix} L/2 & 0 \\ 1 & 0 \\ -L & R \\ 0 & 1 \end{bmatrix} \quad \Delta k = \begin{bmatrix} L/2 & 0 \\ -L/2 & 2R \\ +L & -R \end{bmatrix} \quad \Delta \phi = \begin{bmatrix} 0 & -1 \end{bmatrix}$$

$$K_s = \begin{bmatrix} -m_1 g L/2 & 0 \\ 0 & 0 \end{bmatrix}$$